

JIMI ADEMOLA

Canada

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Education

McGill University

Sep. 2026 - Ongoing

Master's of Science in Computer Science

University of British Columbia

Sep. 2022 – May 2026

*Bachelor of Science in Computer Science (**Honours**) - Dean's Scholar*

Experience

University of British Columbia

Sep 2025 – May 2026

QA Software Developer

Kelowna, BC

- Led deployment of the **LearnCoding** platform beyond university environment, ensuring reliable CI/CD integration and uptime.
- Implemented cybersecurity scanning tools and protocols in coordination with the compliance team to identify vulnerabilities and safeguard user data.
- Developed backend services using PHP and the Laravel framework to support authentication, course delivery, and user management.

Natural Sciences and Engineering Research Council of Canada

May 2025 – Aug. 2025

Undergraduate Research Intern

Kelowna, BC

- Created novel algorithms using statistical weakness to attack and crack various pseudorandom number generators (PRNGs).
- Utilize novel wildfire modelling techniques (GIS) including data scraping, analysis and simulation using R and Python.
- Modelled Pull Request (PR) integration probabilities by evaluating repository metadata with machine learning models in R.

TELUS

May 2023 – Aug 2023

Data & Analytics Intern

Edmonton, AB

- Employ data modelling and visualization tools in Python and Tableau to model user trends throughout company services.
- Evaluate early LLMs to ensure reliable harnesses, compliance and transferable usability between departments.
- Coordinate with cross-functional team within the Slack Workspace to complete tasks using the Agile Kanban workflow.

Projects

Project Artifact Miner | *Python, Electron, FastAPI*

- Developed a static software miner utilizing machine learning models to analyze a user's projects and generate insights
- Produced both a *pip* package for command line use and a front-end powered by Electron
- Implemented parallel computing principles to produce a 300% speedup in file processing time
- Generated 20+ statistics and insights for display on a user's portfolio or resume

GoGuard IPS | *Go, Kali Linux, Ubuntu*

- Engineered a real-time Intrusion Prevention System in Go with a five-stage packet inspection pipeline and automated **iptables** enforcement
- Developed stateful TCP flow reassembly to detect fragmentation-based DPI evasion across split payloads spanning multiple packets
- Implemented MAC-IP binding spoofing detection cross-referenced against the kernel ARP table for attribution

Research Contributions

Beyond the Pull Request: Examining Code Quality in Mature Open Source Projects | (2026) **In preparation**

[In]Secure: Using Statistical Weaknesses to Crack Pseudo-Random Number Generators | (2025)

Technical Skills

Languages: Python, Java, C, HTML/CSS/JavaScript, TypeScript, SQL, R, PHP

Tools: Docker, GCP, Android Studio, Git, Jupyter, Tableau

Technologies/Frameworks: Linux, JUnit, WordPress, MITRE ATT&CK,, React, Astro, Cloudflare Pages